VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JnanaSangama", Belgaum -590014, Karnataka.

LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING

B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU-560019
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CERTIFICATE

B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering

This is to certify that the Lab work entitled **"ANALYSIS AND DESIGN OF ALGORITHMS"** carried out by ANAGHA A PAI (1BM24CS038), who is a bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - **(23CS4PCADA)** work prescribed for the said degree.

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Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

Lab program 1:

Write a program to obtain the Topological ordering of vertices in a given digraph.

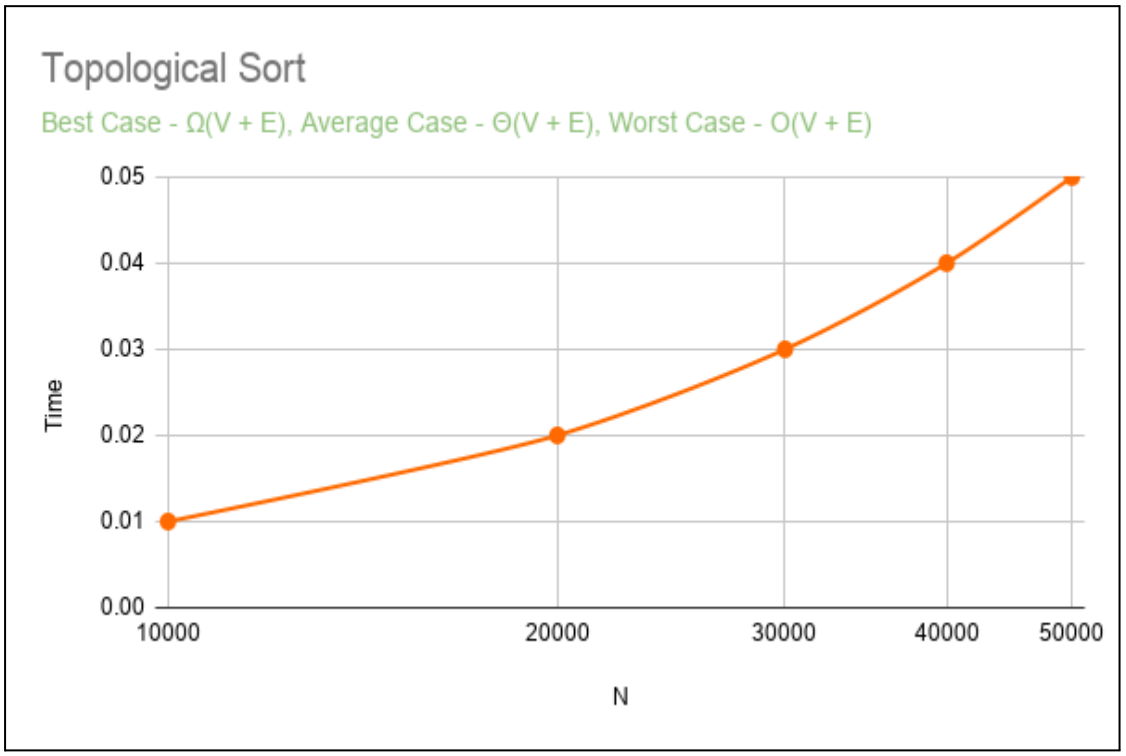
```
#include <stdio.h>

#define MAX 10

int adj[MAX][MAX], visited[MAX], stack[MAX];
int n, top = -1;

void dfs(int v) {
   visited[v] = 1;

   for (int i = 0; i < n; i++) {
      if (adj[v][i] && !visited[i])
         dfs(i);
   }

   stack[++top] = v;
}

void topoSort() {
   for (int i = 0; i < n; i++) {
      if (!visited[i])
         dfs(i);
   }

   printf("Topological Order:\n");
   for (int i = top; i >= 0; i--)
      printf("%d ", stack[i]);
}

int main() {
   printf("Enter number of vertices: ");
   scanf("%d", &n);

   printf("Enter adjacency matrix:\n");
   for (int i = 0; i < n; i++) {
      visited[i] = 0;
      for (int j = 0; j < n; j++) {
         scanf("%d", &adj[i][j]);
      }
```

```
    }

    topoSort();
    return 0;
}
```

Output:

```
Enter number of vertices: 4
Enter adjacency matrix:
0 1 1 0
0 0 0 1
0 0 0 1
0 0 0 0
Topological Order:
0 2 1 3

=== Code Execution Successful ===
```

Graph:



Lab program 4: LEETCODE - Course Schedule

There are a total of numCourses courses you have to take, labeled from 0 to numCourses - 1. You are given an array prerequisites where prerequisites[i] = [a_i, b_i] indicates that you **must** take course b_i first if you want to take course a_i.

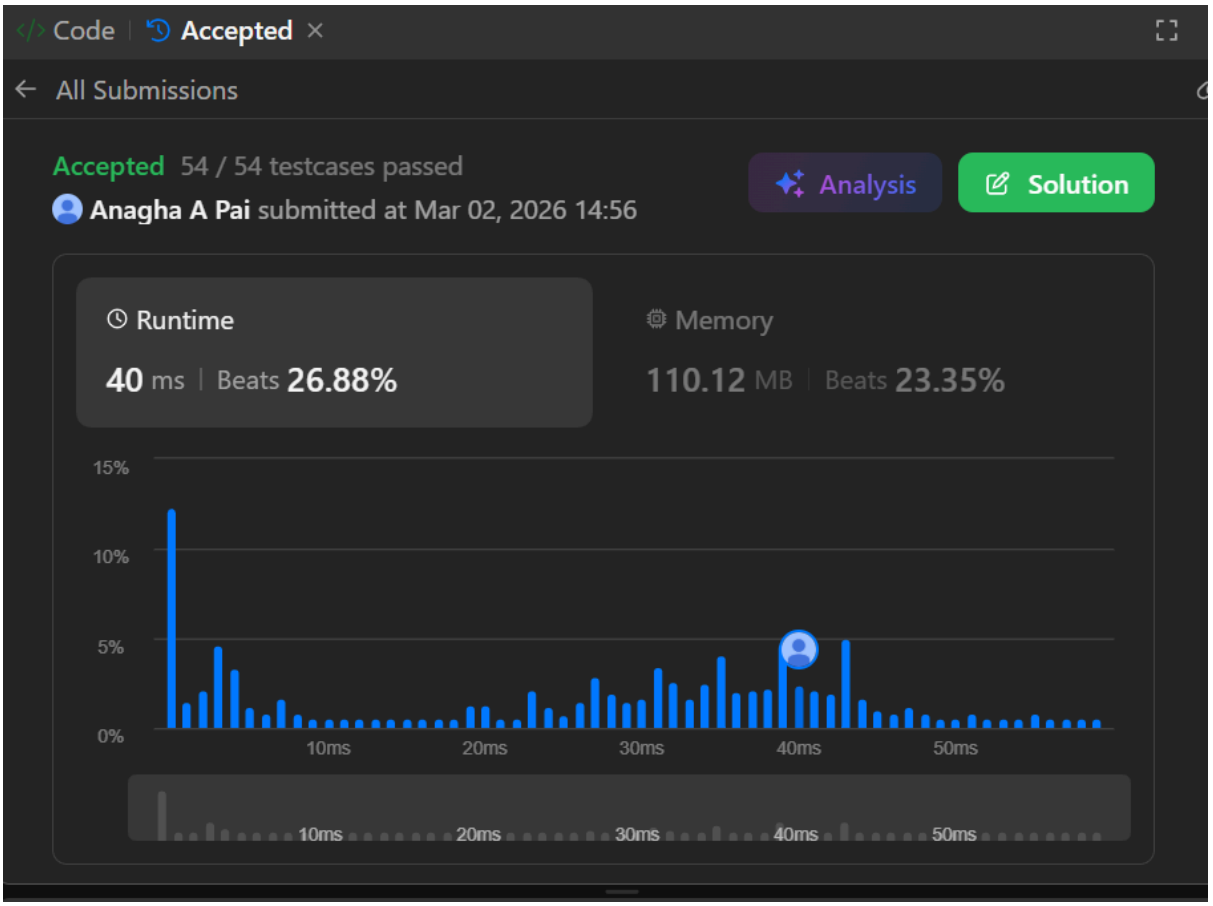
For example, the pair [0, 1], indicates that to take course 0 you have to first take course 1.

Return true if you can finish all courses. Otherwise, return false.

```
bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize,
int* prerequisitesColSize) {
    int* indegree = (int*)calloc(numCourses, sizeof(int));
    int** graph = (int**)malloc(numCourses * sizeof(int*));
    int* graphSize = (int*)calloc(numCourses, sizeof(int));
    for (int i = 0; i < numCourses; i++) {
        graph[i] = (int*)malloc(prerequisitesSize * sizeof(int));
    }
    for (int i = 0; i < prerequisitesSize; i++) {
        int course = prerequisites[i][0];
        int prereq = prerequisites[i][1];

        graph[prereq][graphSize[prereq]++] = course;
        indegree[course]++;
    }
    int* queue = (int*)malloc(numCourses * sizeof(int));
    int front = 0, rear = 0;
    for (int i = 0; i < numCourses; i++) {
        if (indegree[i] == 0)
            queue[rear++] = i;
    }
    int completed = 0;
    while (front < rear) {
        int curr = queue[front++];
        completed++;
        for (int i = 0; i < graphSize[curr]; i++) {
            int next = graph[curr][i];
            indegree[next]--;
            if (indegree[next] == 0)
                queue[rear++] = next;
        }
    }
    free(indegree);
    free(queue);
```

```
    free(graphSize);
    for (int i = 0; i < numCourses; i++)
        free(graph[i]);
    free(graph);
    return completed == numCourses;
}
```

Output:



Lab program 2:

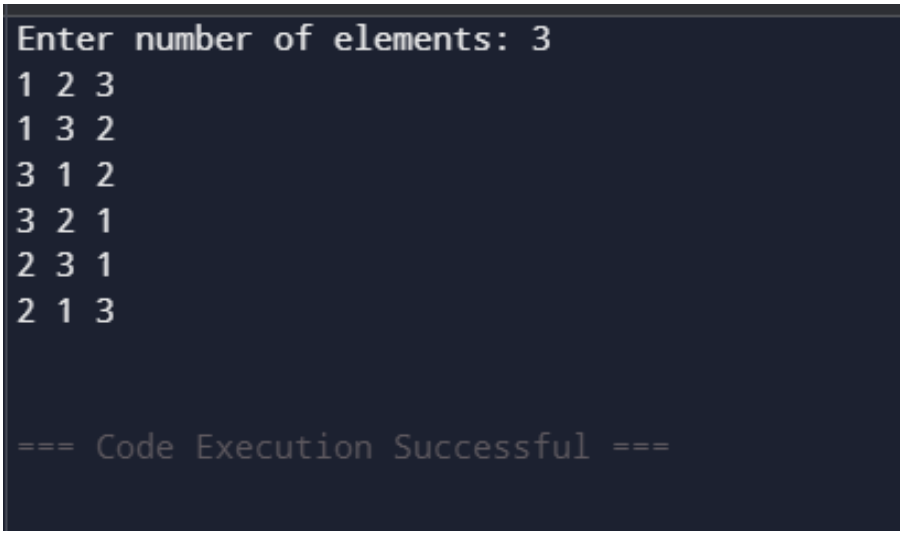
Implement Johnson Trotter algorithm to generate permutations.

```
#include <stdio.h>

#define LEFT -1
#define RIGHT 1
void printPermutation(int perm[], int n) {
   for (int i = 0; i < n; i++)
      printf("%d ", perm[i]);
   printf("\n");
}
int getMobile(int perm[], int dir[], int n) {
   int mobile = 0, mobile_index = -1;

   for (int i = 0; i < n; i++) {
      if (dir[perm[i] - 1] == LEFT && i != 0) {
         if (perm[i] > perm[i - 1] && perm[i] > mobile) {
            mobile = perm[i];
            mobile_index = i;
         }
      }

      if (dir[perm[i] - 1] == RIGHT && i != n - 1) {
         if (perm[i] > perm[i + 1] && perm[i] > mobile) {
            mobile = perm[i];
            mobile_index = i;
         }
      }
   }

   return mobile_index;
}

void swap(int *a, int *b) {
   int temp = *a;
   *a = *b;
   *b = temp;
}
```

```
void johnsonTrotter(int n) {
   int perm[n], dir[n];


   for (int i = 0; i < n; i++) {
      perm[i] = i + 1;
      dir[i] = LEFT;
   }

   printPermutation(perm, n);

   while (1) {
      int mobile_index = getMobile(perm, dir, n);

      if (mobile_index == -1)
         break;

      int mobile = perm[mobile_index];

      // Swap in its direction
      if (dir[mobile - 1] == LEFT)
         swap(&perm[mobile_index], &perm[mobile_index - 1]);
      else
         swap(&perm[mobile_index], &perm[mobile_index + 1]);

      int new_index;
      for (int i = 0; i < n; i++) {
         if (perm[i] == mobile) {
            new_index = i;
            break;
         }
      }

        for (int i = 0; i < n; i++) {
         if (perm[i] > mobile)
            dir[perm[i] - 1] *= -1;
      }

      printPermutation(perm, n);
   }
```

```
}

int main() {
    int n;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    johnsonTrotter(n);

    return 0;
}
```

Output:



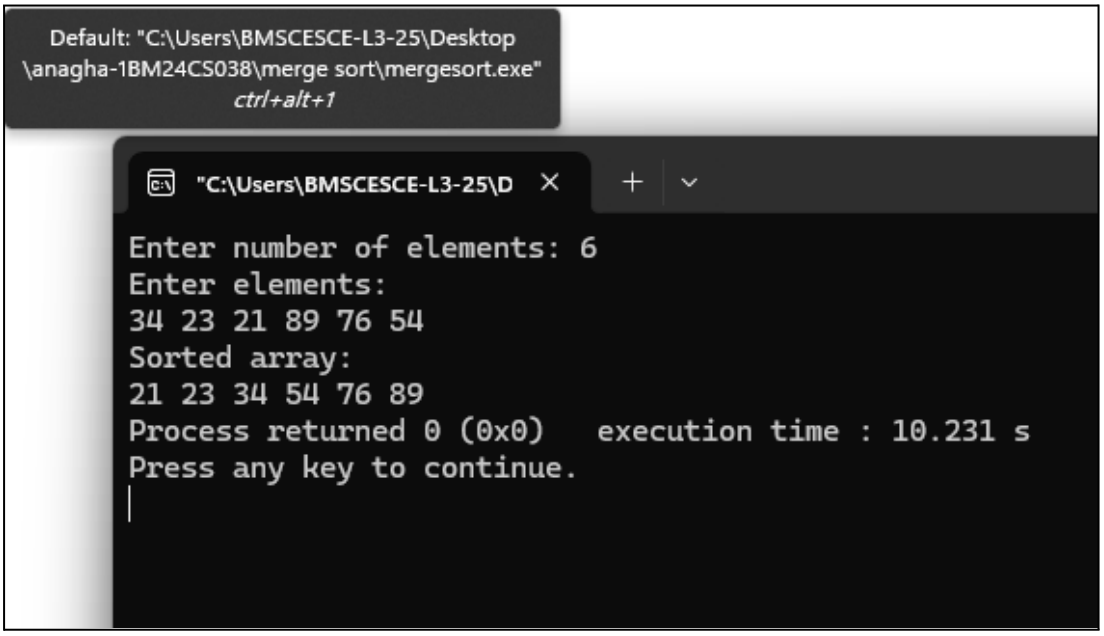
Lab program 3:

Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

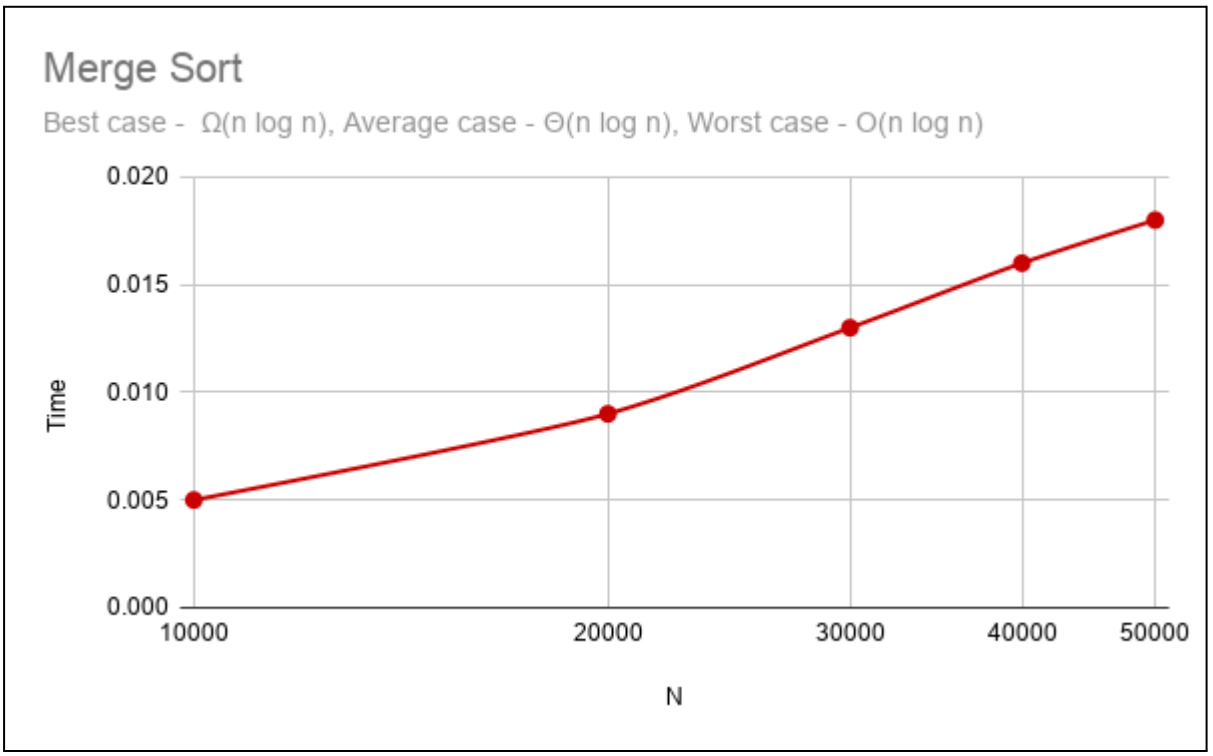
```
#include <stdio.h>
#include <stdlib.h>
void merge(int arr[], int l, int m, int r) {
    int i, j, k;
    int n1 = m - l + 1;
    int n2 = r - m;
    int L[n1], R[n2];

    for (i = 0; i < n1; i++)
        L[i] = arr[l + i];
    for (j = 0; j < n2; j++)
        R[j] = arr[m + 1 + j];

    i = 0; j = 0; k = l;
    while (i < n1 && j < n2) {
        if (L[i] <= R[j]) {
            arr[k] = L[i];
            i++;
        } else {
            arr[k] = R[j];
            j++;
        }
        k++;
    }
    while (i < n1) {
        arr[k++] = L[i++];
    }

    while (j < n2) {
        arr[k++] = R[j++];
    }
}
void mergeSort(int arr[], int l, int r) {
    if (l < r) {
        int m = (l + r) / 2;
```

```
        mergeSort(arr, l, m);
        mergeSort(arr, m + 1, r);

        merge(arr, l, m, r);
    }
}

int main() {
    int n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int *arr = (int *)malloc(n * sizeof(int));

    printf("Enter elements:\n");
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    mergeSort(arr, 0, n - 1);

    printf("Sorted array:\n");
    for (i = 0; i < n; i++)
        printf("%d ", arr[i]);

    free(arr);
    return 0;
}
```

Output:



Graph:



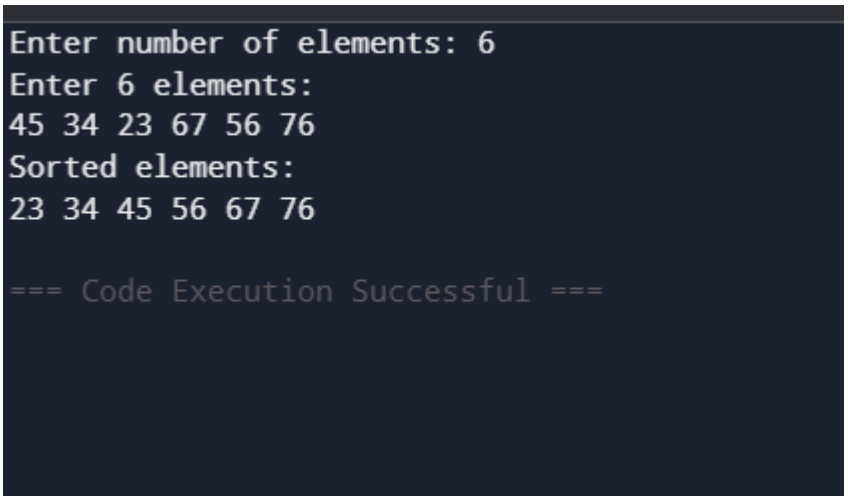
Lab program 4:

Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

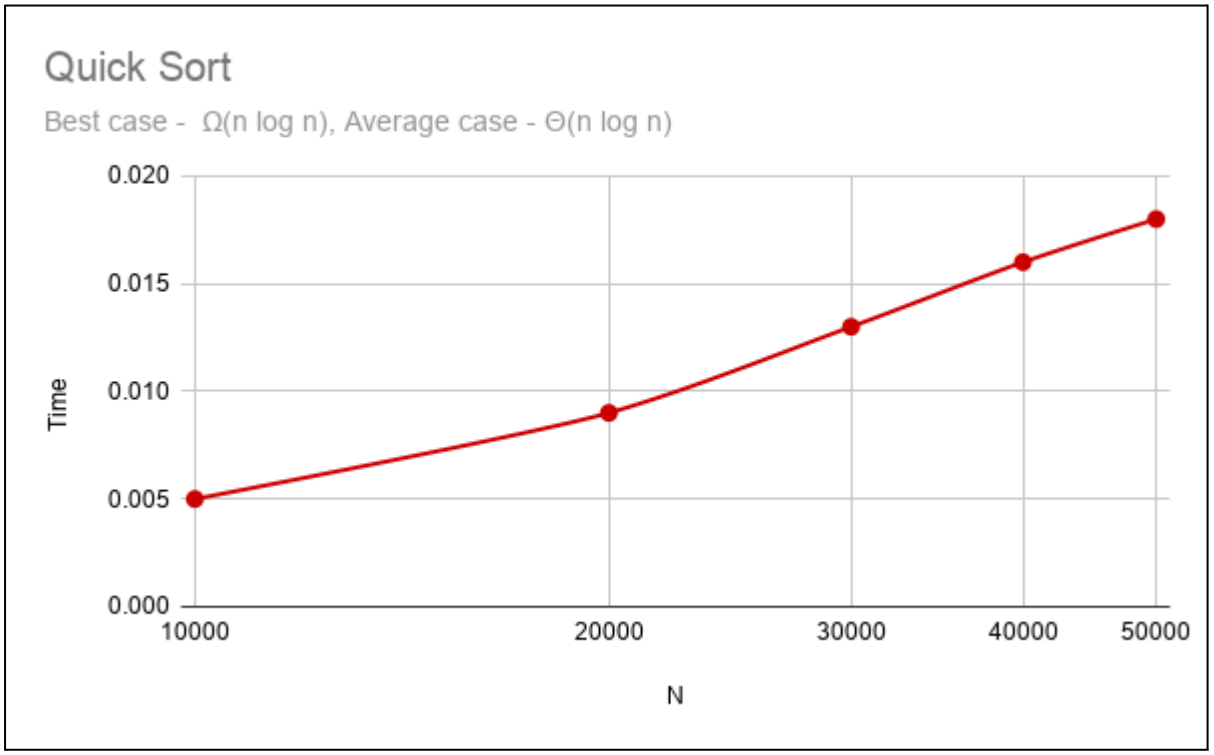
```
#include <stdio.h>

// Swap function
void swap(int *a, int *b) {
   int temp = *a;
   *a = *b;
   *b = temp;
}

int partition(int arr[], int low, int high) {
   int pivot = arr[low];   // first element as pivot
   int i = low + 1;
   int j = high;

   while (1) {
      while (i <= high && arr[i] <= pivot)
         i++;

      while (arr[j] > pivot)
         j--;

      if (i >= j)
         break;

      swap(&arr[i], &arr[j]);
   }

   swap(&arr[low], &arr[j]);
   return j;
}
void quickSort(int arr[], int low, int high) {
   if (low < high) {
      int p = partition(arr, low, high);
      quickSort(arr, low, p - 1);
      quickSort(arr, p + 1, high);
```

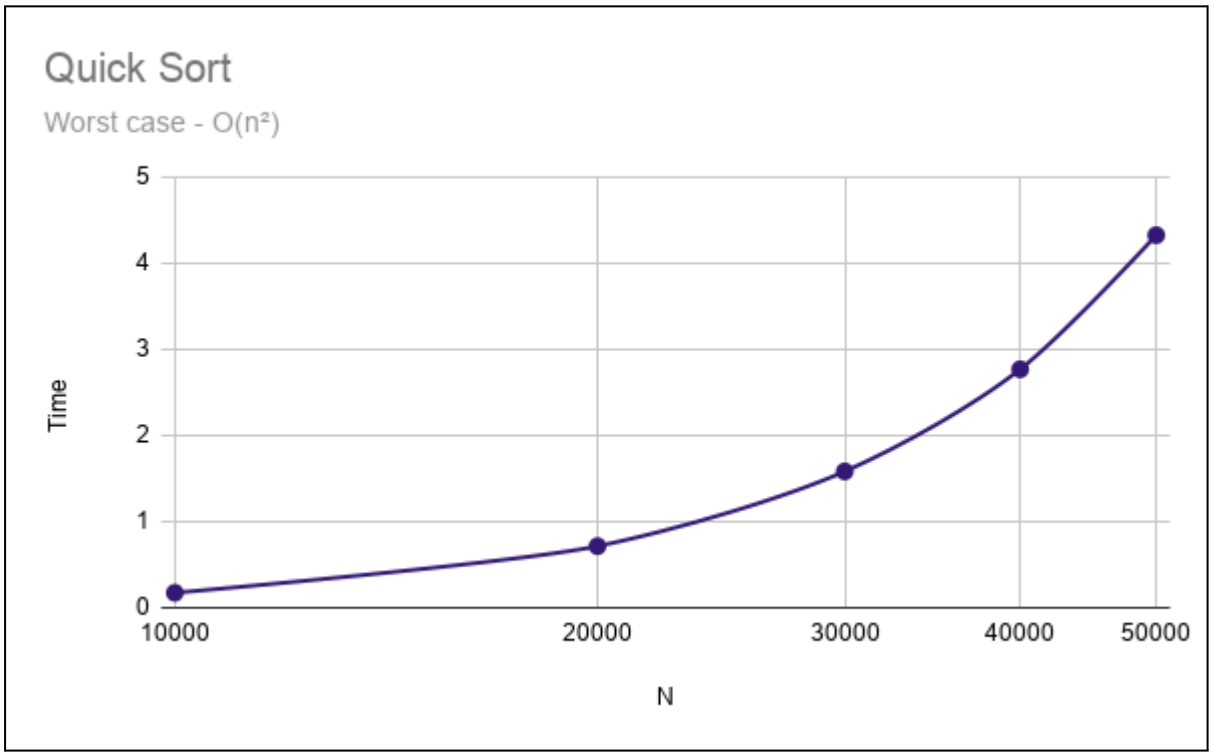
```
    }
}

int main() {
    int n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    quickSort(arr, 0, n - 1);

    printf("Sorted elements:\n");
    for (i = 0; i < n; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

Output:



Graph:


Quick Sort
Best case - Ω(n log n), Average case - Θ(n log n)
0.020
0.015
0.010
0.005
0.000
Time
10000
20000
30000
40000
50000
N


Quick Sort
Worst case - O(n²)
5
4
3
2
1
0
Time
10000
20000
30000
40000
50000
N

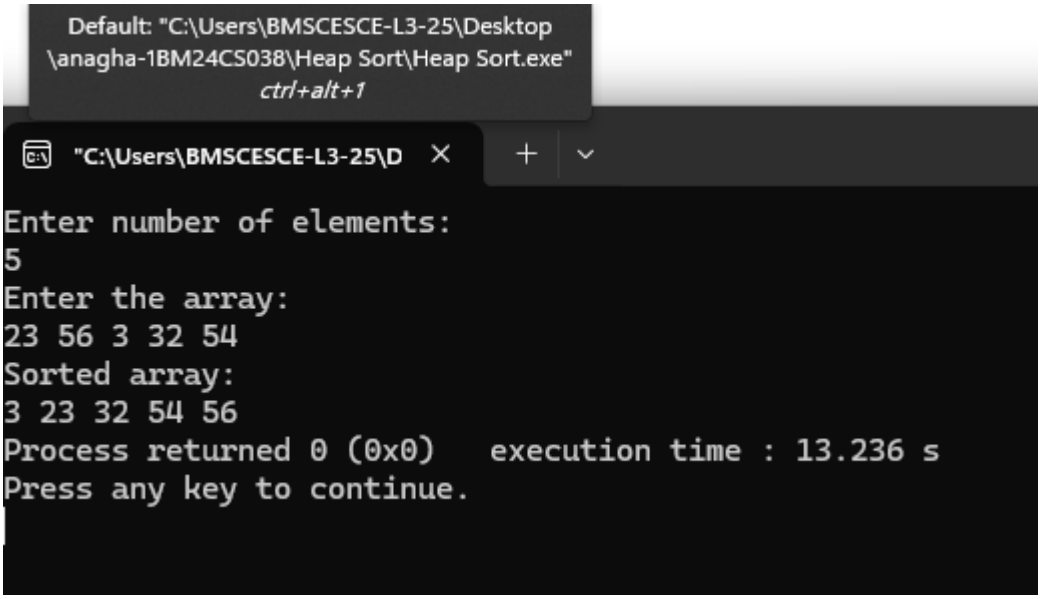
Lab program 5:

Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

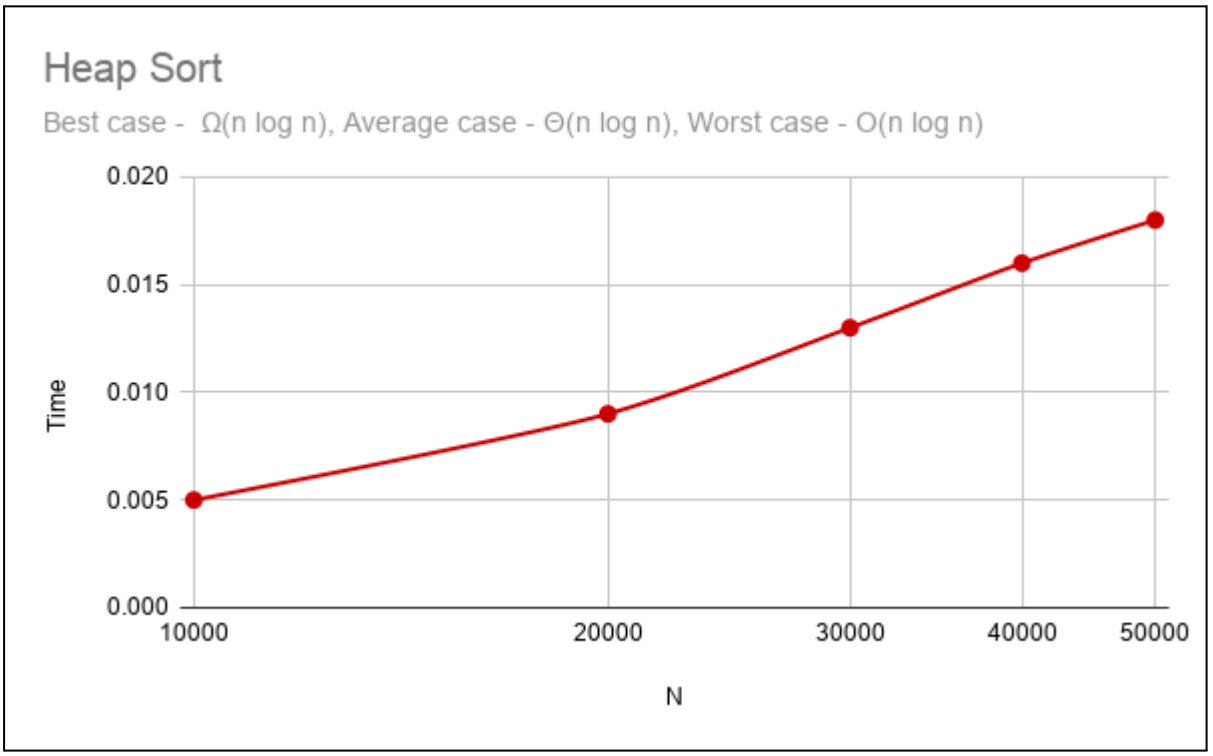
```
#include<stdio.h>

void heapify(int a[], int n, int p)
{
   int c, item;
   item = a[p];
   c = 2 * p + 1;
   while(c < n)
   {

      if(c + 1 < n)
      {
         if(a[c] < a[c + 1])
         {
            c++;
         }
      }
      if(item < a[c])
      {
         a[p] = a[c];
         p = c;
         c = 2 * p + 1;
      }
      else
      {
         break;
      }
   }

   a[p] = item;
}

void build_heap(int a[], int n)
{
   int i;

   for(i = (n - 1) / 2; i >= 0; i--)
   {
```

```
        heapify(a, n, i);
    }
}

void heap_sort(int a[], int n)
{
    int i, temp;
    build_heap(a, n);

    for(i = n - 1; i > 0; i--)
    {
        temp = a[0];
        a[0] = a[i];
        a[i] = temp;

        heapify(a, i, 0);
    }
}

int main()
{
    int a[100], n;
    printf("Enter number of elements:\n");
    scanf("%d", &n);
    printf("Enter the array:\n");

    for(int i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    heap_sort(a, n);
    printf("Sorted array:\n");
    for(int i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }

    return 0;
}
```

Output:



Graph:



Lab program 6:

Implement 0/1 Knapsack problem using dynamic programming.

```
#include <stdio.h>
int max(int a, int b) {
   return (a > b) ? a : b;
}
int knapsack(int W, int wt[], int val[], int n) {
   int dp[n+1][W+1];
   for (int i = 0; i <= n; i++) {
      for (int w = 0; w <= W; w++) {
         if (i == 0 || w == 0)
            dp[i][w] = 0;
         else if (wt[i-1] <= w)
            dp[i][w] = max(val[i-1] + dp[i-1][w - wt[i-1]],
                           dp[i-1][w]);
         else
            dp[i][w] = dp[i-1][w];
      }
   }
   return dp[n][W];
}
int main() {
   int n, W;
   printf("Enter number of items: ");
   scanf("%d", &n);
   int val[n], wt[n];
   printf("Enter values of items:\n");
   for (int i = 0; i < n; i++) {
```

```
        scanf("%d", &val[i]);
    }
    printf("Enter weights of items:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &wt[i]);
    }
    printf("Enter capacity of knapsack: ");
    scanf("%d", &W);
    printf("Maximum value = %d\n", knapsack(W, wt, val, n));
    return 0;
}
```

Output:

```
Enter number of items: 4
Enter values of items:
3 4 5 6
Enter weights of items:
1 2 3 5
Enter capacity of knapsack: 8
Maximum value = 13


=== Code Execution Successful ===
```

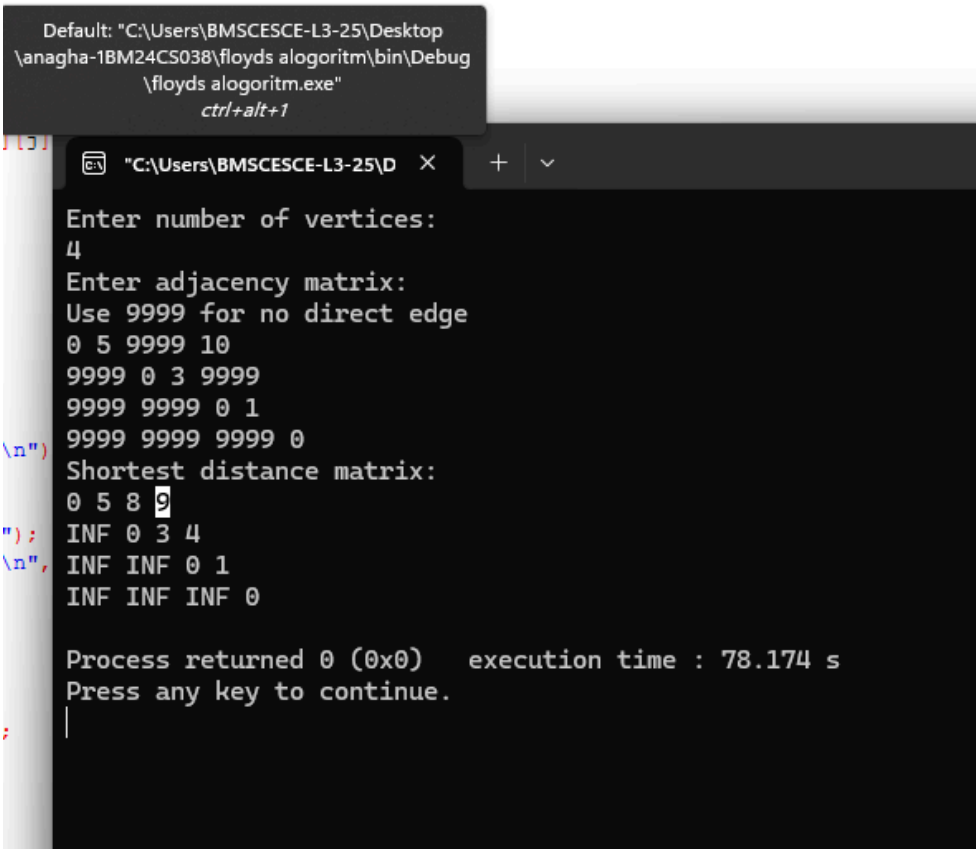
Lab program 7:

Implement All Pair Shortest paths problem using Floyd's algorithm.

```
#include<stdio.h>
#define INF 9999

void floydWarshall(int graph[10][10], int n)
{
   int i, j, k;

   // Updating shortest distances
   for(k = 0; k < n; k++)
   {
      for(i = 0; i < n; i++)
      {
         for(j = 0; j < n; j++)
         {
            if(graph[i][k] + graph[k][j] < graph[i][j])
            {
               graph[i][j] = graph[i][k] + graph[k][j];
            }
         }
      }
   }
   printf("Shortest distance matrix:\n");
   for(i = 0; i < n; i++)
   {
      for(j = 0; j < n; j++)
      {
         if(graph[i][j] == INF)
            printf("INF ");
         else
            printf("%d ", graph[i][j]);
      }
      printf("\n");
   }
}

int main()
{
```

```
    int n, i, j;
    int graph[10][10];
    printf("Enter number of vertices:\n");
    scanf("%d", &n);
    printf("Enter adjacency matrix:\n");
    printf("Use %d for no direct edge\n", INF);
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &graph[i][j]);
        }
    }
    floydWarshall(graph, n);
    return 0;
}
```

Output:



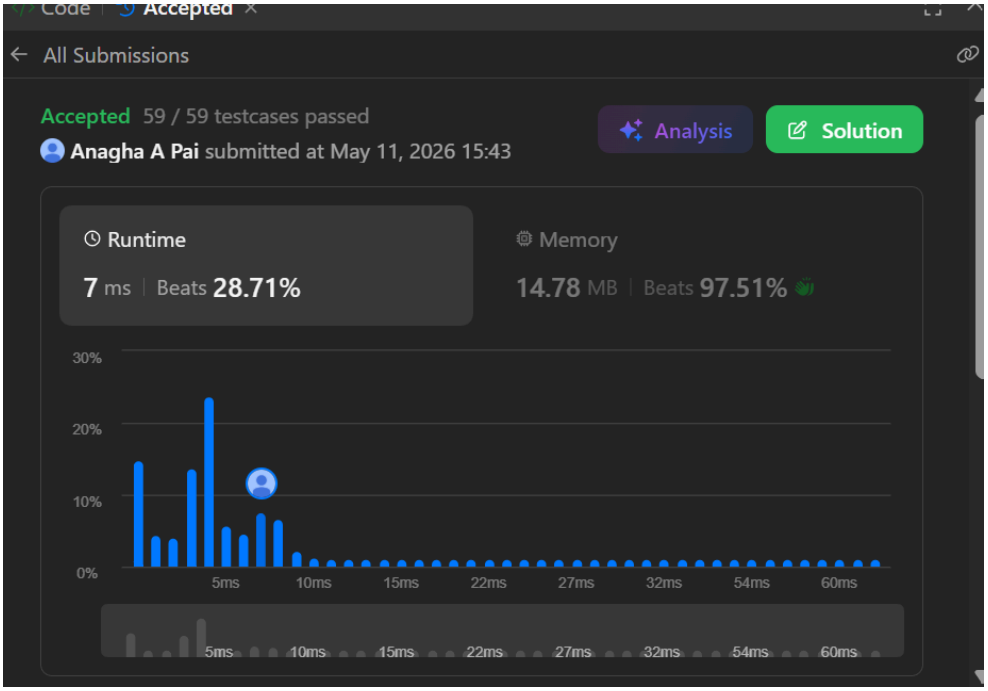
Lab program 10: LEETCODE - Find the duplicate number

Given an array of integers nums containing n + 1 integers where each integer is in the range [1, n] inclusive.

There is only **one repeated number** in nums, return *this repeated number*.

```c
int findDuplicate(int* nums, int numsSize) {
    int slow = nums[0];
    int fast = nums[0];
       do {
        slow = nums[slow];
        fast = nums[nums[fast]];
    } while (slow != fast);
      slow = nums[0];
    while (slow != fast) {
        slow = nums[slow];
        fast = nums[fast];
    }
    return slow;
}
```

Output:



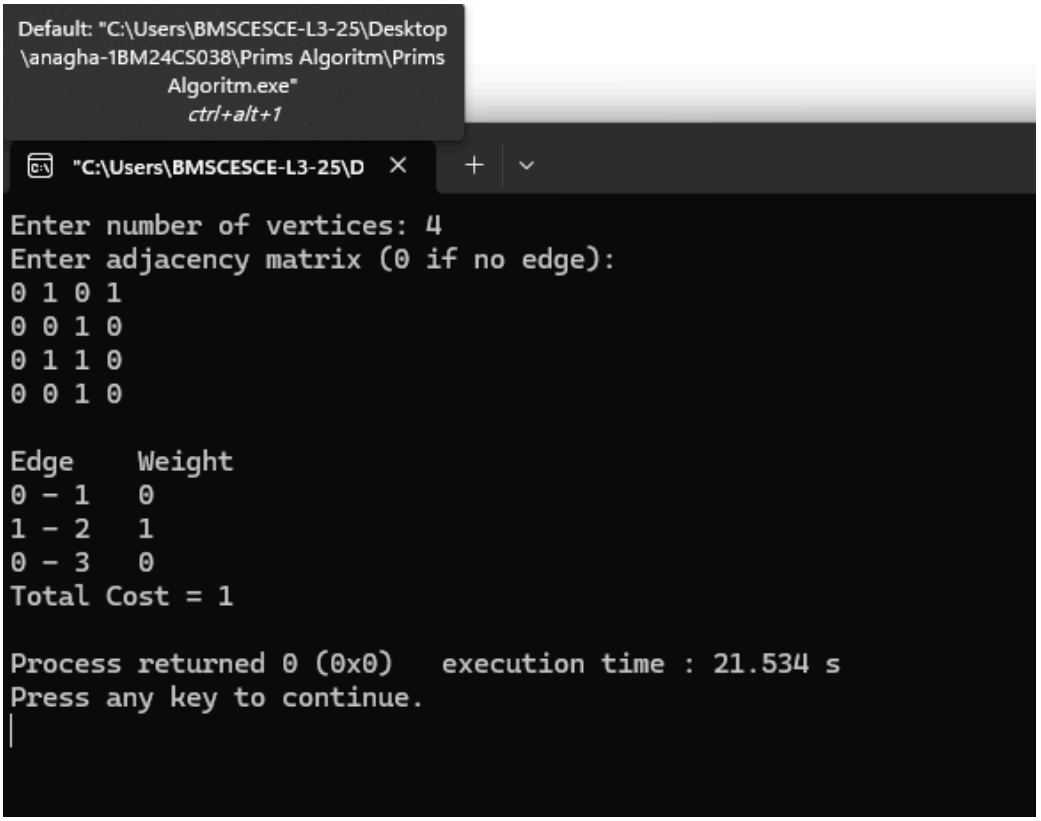
Lab program 8a:

Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.

```
#include <stdio.h>
#include <limits.h>

#define m 100

int V;

int minKey(int key[], int mstSet[]) {
   int min = INT_MAX, min_index;

   for (int v = 0; v < V; v++) {
      if (mstSet[v] == 0 && key[v] < min) {
         min = key[v];
         min_index = v;
      }
   }
   return min_index;
}

void printMST(int parent[], int graph[m][m]) {
   int cost = 0;
   printf("\nEdge \tWeight\n");
   for (int i = 1; i < V; i++) {
      printf("%d - %d \t%d\n", parent[i], i, graph[i][parent[i]]);
      cost += graph[i][parent[i]];
   }
   printf("Total Cost = %d\n", cost);
}

void primMST(int graph[m][m]) {
   int parent[m], key[m], mstSet[m];

   for (int i = 0; i < V; i++) {
      key[i] = INT_MAX;
      mstSet[i] = 0;
   }
```

```
    key[0] = 0;
    parent[0] = -1;

    for (int count = 0; count < V - 1; count++) {
        int u = minKey(key, mstSet);
        mstSet[u] = 1;

        for (int v = 0; v < V; v++) {
            if (graph[u][v] && mstSet[v] == 0 && graph[u][v] < key[v]) {
                parent[v] = u;
                key[v] = graph[u][v];
            }
        }
    }

    printMST(parent, graph);
}

int main() {
    int graph[m][m];

    printf("Enter number of vertices: ");
    scanf("%d", &V);

    printf("Enter adjacency matrix (0 if no edge):\n");
    for (int i = 0; i < V; i++) {
        for (int j = 0; j < V; j++) {
            scanf("%d", &graph[i][j]);
        }
    }

    primMST(graph);

    return 0;
}
```

Output:


Default: "C:\Users\BMSCESCE-L3-25\Desktop
\anagha-1BM24CS038\Prims Algoritm\Prims
Algoritm.exe"
ctrl+alt+1
"C:\Users\BMSCESCE-L3-25\D
Enter number of vertices: 4
Enter adjacency matrix (0 if no edge):
0 1 0 1
0 0 1 0
0 1 1 0
0 0 1 0
Edge Weight
0 - 1 0
1 - 2 1
0 - 3 0
Total Cost = 1
Process returned 0 (0x0) execution time : 21.534 s
Press any key to continue.

Lab program 8b:

Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

```
#include<stdio.h>
#include<string.h>
#include <stdio.h>
#include <stdlib.h>
#define MAX 100

struct Edge {
   int src, dest, weight;
};

int parent[MAX];

int find(int i) {
   while (parent[i] != i)
      i = parent[i];
   return i;
}

void unionSet(int u, int v) {
   int u_root = find(u);
   int v_root = find(v);
   parent[u_root] = v_root;
}

int compare(const void* a, const void* b) {
   return ((struct Edge*)a)->weight - ((struct Edge*)b)->weight;
}

int main() {
   int V, E;

   printf("Enter number of vertices: ");
   scanf("%d", &V);

   printf("Enter number of edges: ");
   scanf("%d", &E);
```

```
    struct Edge edges[MAX];

    printf("Enter edges (src dest weight):\n");
    for (int i = 0; i < E; i++) {
        scanf("%d %d %d", &edges[i].src, &edges[i].dest, &edges[i].weight);
    }

    for (int i = 0; i < V; i++)
        parent[i] = i;

    qsort(edges, E, sizeof(edges[0]), compare);

    int count = 0, i = 0, cost = 0;

    printf("\nEdge \tWeight\n");

    while (count < V - 1 && i < E) {
        struct Edge next = edges[i++];

        int x = find(next.src);
        int y = find(next.dest);

        if (x != y) {
            printf("%d - %d \t%d\n", next.src, next.dest, next.weight);
            cost += next.weight;
            unionSet(x, y);
            count++;
        }
    }

    printf("Total Cost = %d\n", cost);

    return 0;
}
```

Output:

```
Enter number of vertices: 4
Enter number of edges: 5
Enter edges (src dest weight):
0 1 10
0 2 6
0 3 5
1 3 15
2 3 4

Edge    Weight
2 - 3   4
0 - 3   5
0 - 1   10
Total Cost = 19


=== Code Execution Successful ===
```

Lab program 9:

Implement fractional knapsack problem using Greedy technique.

```
#include<stdio.h>
void main()
{

   float p[100],w[100],x[100],n,maxcap,total=0.0,temp1,temp2;
   printf("enter the  number of elements\n");
   scanf("%f",&n);
   float rc=0;
   printf("Enter the values\n");
   for(int i=0;i<n;i++)
   {
      scanf("%f",&p[i]);
      scanf("%f",&w[i]);
      x[i]=0.0;

   }
   printf("enter the maximum capacity\n");
   scanf("%f",&maxcap);
   rc=maxcap;
   for(int i=0;i<n;i++)
   {
      for(int k=0;k<n-1;k++)
      {
         if(p[i]/w[i]>p[k]/w[k])
         {
            temp1=p[i];
            p[i]=p[k];
            p[k]=temp1;
            temp2=w[i];
            w[i]=w[k];
            w[k]=temp2;
         }
      }
   }
   int j;
   for( j=0;j<n;j++)
   {

      if(w[j]<rc)
```

```
        {
            rc-=w[j];
            x[j]=1.0;
        }
        else
            break;
    }
    if(j<n)
    {
        x[j]=rc/w[j];
    }
    for(int i=0;i<n;i++)
    {
        total=total+(x[i]*p[i]);
    }
    printf("the total profit: %f",total);
}
```

Output:

```
enter the  number of elements
5
Enter the values
4 5 6 7 3
6 7 4 5 3
enter the maximum capacity
8
the total profit: 12.857143

=== Code Exited With Errors ===
```

Lab program 10:

From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.

```
#include <stdio.h>
#define INF 99999
int minDistance(int dist[], int visited[], int n) {
   int min = INF, min_index = -1;
   for (int v = 0; v < n; v++) {
      if (!visited[v] && dist[v] <= min) {
         min = dist[v];
         min_index = v;
      }
   }
   return min_index;
}
void dijkstra(int n, int graph[n][n], int src) {
   int dist[n], visited[n];

   for (int i = 0; i < n; i++) {
      dist[i] = INF;
      visited[i] = 0;
   }

   dist[src] = 0;
   for (int count = 0; count < n - 1; count++) {
      int u = minDistance(dist, visited, n);
      if (u == -1) break;
      visited[u] = 1;
      for (int v = 0; v < n; v++) {
         if (!visited[v] && graph[u][v] &&
            dist[u] != INF &&
            dist[u] + graph[u][v] < dist[v]) {

            dist[v] = dist[u] + graph[u][v];
         }
      }
   }

   printf("Vertex\tDistance from Source\n");
   for (int i = 0; i < n; i++) {
      printf("%d\t%d\n", i, dist[i]);
```

```
    }
}

int main() {
    int n;
    printf("Enter the number of vertices of the graph:\n");
    scanf("%d", &n);

    int graph[n][n];

    printf("Enter the adjacency matrix values:\n");
    for(int i = 0; i < n; i++) {
        for(int j = 0; j < n; j++) {
            scanf("%d", &graph[i][j]);
        }
    }

    dijkstra(n, graph, 0);
    return 0;
}
```

Output:

```
Enter the number of vertices of the graph:
5
Enter the adjacency matrix values:
0   10   0   30   100
0    0  50    0     0
0    0   0    0    10
0    0  20    0    60
0    0   0    0     0
Vertex  Distance from Source
0   0
1   10
2   50
3   30
4   60
```

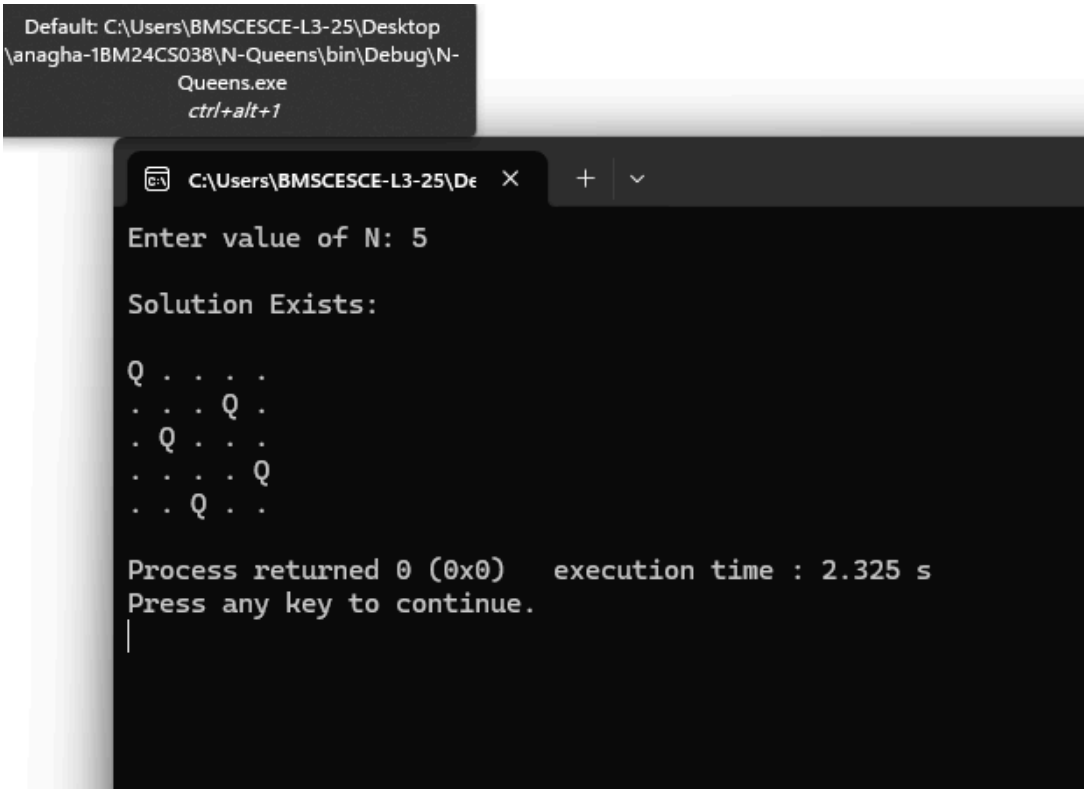
Lab program 11:

Implement "N-Queens Problem" using Backtracking.

```
#include <stdio.h>

#define MAX 20

int board[MAX][MAX];
int N;
int isSafe(int row, int col) {
   int i, j;

   // Check left side of current row
   for (i = 0; i < col; i++) {
      if (board[row][i])
         return 0;
   }
   for (i = row, j = col; i >= 0 && j >= 0; i--, j--) {
      if (board[i][j])
         return 0;
   }
   for (i = row, j = col; i < N && j >= 0; i++, j--) {
      if (board[i][j])
         return 0;
   }

   return 1;
}
int solveNQ(int col) {

   // If all queens are placed
   if (col >= N)
      return 1;

   int i;
   for (i = 0; i < N; i++) {

      // Check if safe
      if (isSafe(i, col)) {

         // Place queen
         board[i][col] = 1;
```

```
            if (solveNQ(col + 1))
                return 1;
            board[i][col] = 0;
        }
    }

    return 0;
}
void printBoard() {
    int i, j;

    printf("\nSolution Exists:\n\n");

    for (i = 0; i < N; i++) {
        for (j = 0; j < N; j++) {

            if (board[i][j])
                printf("Q ");
            else
                printf(". ");
        }
        printf("\n");
    }
}

int main() {

    printf("Enter value of N: ");
    scanf("%d", &N);

    if (solveNQ(0))
        printBoard();
    else
        printf("No Solution Exists");

    return 0;
}
```

Output:


Default: C:\Users\BMSCESCE-L3-25\Desktop\anagha-1BM24CS038\N-Queens\bin\Debug\N-Queens.exe
ctrl+alt+1
C:\Users\BMSCESCE-L3-25\De
Enter value of N: 5
Solution Exists:
Q . . . .
. . . Q .
. Q . . .
. . . . Q
. . Q . .
Process returned 0 (0x0) execution time : 2.325 s
Press any key to continue.

Lab program : LEETCODE - Container with most water

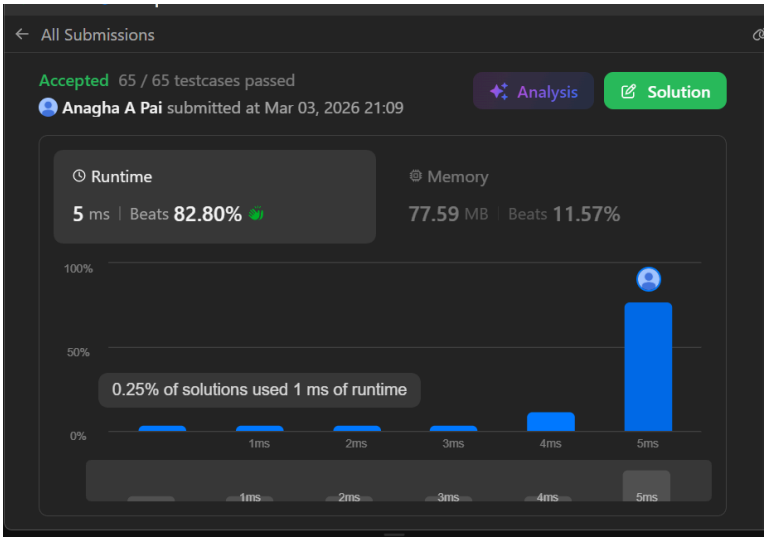
You are given an integer array height of length n. There are n vertical lines drawn such that the two endpoints of the i^{th} line are (i, 0) and (i, height[i]).

Find two lines that together with the x-axis form a container, such that the container contains the most water.

Return *the maximum amount of water a container can store*.

```
int maxArea(int* height, int heightSize) {
    int left = 0;
    int right = heightSize - 1;
    int maxWater = 0;
    while (left < right) {
        int h = (height[left] < height[right]) ? height[left] :
height[right];
        int width = right - left;
        int area = h * width;
        if (area > maxWater)
            maxWater = area;
        if (height[left] < height[right])
            left++;
        else
            right--;
    }
    return maxWater;
}
```

Output:



Lab program : LEETCODE - First and Last position of element in a sorted array

Given an array of integers nums sorted in non-decreasing order, find the starting and ending position of a given target value.

If the target is not found in the array, return [-1, -1].

```
int findFirst(int* nums, int numsSize, int target){
    int left = 0, right = numsSize - 1;
    int ans = -1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        if (nums[mid] == target) {
            ans = mid;
            right = mid - 1;
        } else if (nums[mid] < target) {
            left = mid + 1;
        } else {
            right = mid - 1;
        }
    }
   return ans;
}
int findLast(int* nums, int numsSize, int target) {
    int left = 0, right = numsSize - 1;
    int ans = -1;
    while (left <= right) {
        int mid = left + (right - left) / 2;
        if (nums[mid] == target) {
            ans = mid;
```

```
            left = mid + 1;
        } else if (nums[mid] < target) {
            left = mid + 1;
        } else {
            right = mid - 1;
        }
    }
    return ans;
}
int* searchRange(int* nums, int numsSize, int target, int* returnSize) {
    *returnSize = 2;
    int* result = (int*)malloc(2 * sizeof(int));
    result[0] = findFirst(nums, numsSize, target);
    result[1] = findLast(nums, numsSize, target);

    return result;
}
```

Output: